

NSF GRFP Personal Data Pack – JC (Jacob Vaught)

Updated 11/11/2025

1) Program & Status

- **Current status:** 1st-year M.S. at University of South Carolina (UofSC); fellowship period begins **2nd year** of M.S.
- **Advisor:** Dr. Yi Wang (CS/ML/CV background; team includes strong camera expertise and several radar-capable researchers).
- **Primary goal with advisor:** Develop **automatic image labeling** using radar as a cue; prerequisite is **accurate object labeling in radar** (this is the requested funding). Longer term: build **autonomous boats** for scouting new areas.
- **Why this program/advisor fits:** Team culture, continuity from undergrad project, and active agency/industry collaboration rhythms.

2) Team & Meetings Cadence

- 1:1 w/ PI (Dr. Wang): **weekly**
- Undergrads (UGs): **weekly**, rotating presentations
- Radar team: **biweekly**
- CV/ML team: **weekly**, present every 4 weeks
- Hardware team (cameras, etc.): **weekly**, present every 3 weeks
- Whole lab: **weekly**, present **~annually**
- Navy/agency: **quarterly**, present
- Industry partner (Integer LLC): **weekly**, present **once/semester (~3×/yr)**

3) Research Theme (High-Level)

- **Thesis area:** Physics-aware radar–camera perception with radar-guided auto-labeling.
- **Motivation:** Wireless perception is “fun” and compelling; maritime is a natural, challenging testbed. Problem importance framed around safety, autonomy, and national security.

4) Dataset Summary

- **Current size:** ~200k radar sweeps (~56 hours)
- **Locations:** Mostly **SC lakes**, some **Charleston**, some **VA Hampton Roads**
- **Conditions:** ~20% **night** / 80% **day**; ~5% bad weather (remaining good weather)
- **Collection platforms:** Lab boat + **MOKAI**, both GPS-tracked
- **Scene types:** Littoral/coastal & inland lakes; focus on small maritime objects (boats, buoys), shoreline/land discrimination

5) Ground Truth & Quality Control

- **GT sources:** Camera as cross-check; GPS tracks; manual labeling
- **QC policy:** Double-labeling; camera-backed verification; (proposed) 10% double-label consensus + weekly blind spot-checks; privacy steps (blur faces/vessel names; consider location jitter)

6) Baselines & Metrics

- **Baseline: Commercial baseline** (*Tocaro Blue*) — public naming as “commercial baseline” with (*Tocaro Blue*) in parentheses
- **Quoted baseline figures:** ~93% land discrimination; ~74% boat vs. buoy (metric unspecified in source; you prefer **accuracy** as headline)
- **Primary metrics:** Accuracy (headline), **AP_{small}**, class-balanced **F1**, **recall at low FAR**, **edge latency**
- **Label efficiency target:** Match/beat baseline with **≤10%** of labels

7) Research Aims

Aim 1 — Radar-only self-learning (MAE). - Inputs: **Range–azimuth** (no Doppler) - Pretext: Masked spatial blocks (consider polar-aware masking) - Backbone: **Custom lightweight ViT** (Tiny/Small variants for ablation) - Detector: Lightweight head atop frozen backbone; then limited fine-tuning - **Targets:** ≥98% accuracy (land vs. object) and ≥95% accuracy (boat vs. buoy); **+5 AP_{small}** vs. baseline; label usage ≤10% - **Fallbacks:** Contrastive radar pretext; short temporal windows (2–4 sweeps); semi-supervised fine-tuning with pseudo-labels

Aim 2 — Fusion with physics-aware consistency. - Fusion: **Mid-level** (radar encoder + image encoder with cross-attention) - Physics priors in loss: (a) **Feature-space similarity** for matched radar–image regions; (b) **Range/Bearing → image search** overlap encouragement; (c) small-object queries + oversampling - Robustness probes: Night/fog/rain subsets; shoreline clutter; fast movers

Aim 3 — Deployable models & auto-labeling loop. - Edge target: **Jetson Orin**; ≤100 ms end-to-end; ≥30 FPS (stretch 60 FPS) - Acceptable accuracy drop from non-compressed model: ≤3% (land vs. object), ≤7% (boat vs. buoy) - Compression sequence: Auto-label w/ large model → train compact supervised model → QAT → distillation → TensorRT (as needed) - Auto-label curation: Confidence-weighted selection; radar-gated acceptance; ~5% human audit per batch; record “known pitfalls”

Note on timeline: You prefer **no timeline** in the NSF statement, though earlier internal notes referenced: Fall (baseline+dataset), Spring (MAE + fusion prototype), Summer (deploy + field trials + first public benchmark slice).

8) Risk Register

- **MAE underperforms** on radar (largest risk) → Switch to contrastive/temporal pretexts; increase labeled seed set; semi-supervised fine-tuning
- **Motion/registration artifacts** → Ego-motion compensation; better sync; test temporal windows
- **Label scarcity / QC drift** → Double-label key slices; radar-gated acceptance; maintain auditing

9) Deploy/Performance Targets (Edge)

- **Latency:** ≤ 100 ms
- **Throughput:** 30 FPS minimum; 60 FPS stretch
- **Tolerated accuracy drop:** $\leq 3\%$ (land vs. object), $\leq 7\%$ (boat vs. buoy)

10) Publications, Talks, Artifacts

Published/accepted - Springer chapter: *Modal Analysis using a UAV-deployable Wireless Sensor Network* (co-author); contribution: built FEA validation aligning sensor design with simulated modes

In preparation (Jan submissions; lead author, mentoring UGs): 1. **High-Dynamic-Range Radar via Clipping-Aware Multi-Gain Fusion** 2. **Adaptive Gain Selection for Radar via Entropy Maximization** 3. **Pulse-Diverse HDR Imaging for Robust Radar Targeting**

Talks - QPR ($\approx$ mid-Sept): “Seeing in Binary: Resolving Details from Low-Bit Sensors”
(~ 100 attendees; sponsors + non-technical stakeholders)

Open-source (current public repos) - Radar labeling tool: <https://github.com/j-vaught/radar-labeler> - EMCH368 ML teaching lab materials: https://github.com/j-vaught/Machine-Learning_EMCH368 - Misc repos: <https://github.com/j-vaught/Java-Chess-Game> | <https://github.com/j-vaught/MO-CAP>

11) Partners & Collaborations

- **Agency: NIWC** (data-sharing offer for grad research)
- **Potential:** NOAA, USCG
- **Industry: Unnamed industry partner** (keep anonymous in public docs)

12) Resources & Facilities

- **Sensing:** Furuno & Simrad radomes; RGB + thermal + IR cameras
- **Compute:** Jetson Orin; GPU servers; **Hyperion** HPC cluster
- **Platforms:** Lab boat; **MOKAI**
- **Access:** “Anything I need to use in the lab I have access to.”

13) Skills & Tooling

- **Software/ML:** CUDA; PyTorch; ONNX; OpenCV; Linux; Git; some Docker; C++ for low-level radar comms; radar simulation (custom + MATLAB)
- **Hardware:** FPGAs; flexible PCB/3D-printed circuits; heart-rate sensors; Arduinos; Raspberry Pis; structural sensors; metal 3D printers (LPBFAM)

14) Project History (Selected)

- **Radar control & GUI (senior):** Built end-to-end pipeline from radome to computer; full settings control; intuitive GUI; backbone of current research
- **Radar ML + camera (current):** Building MAE-based radar pretraining; fusion with physics-aware consistency; (no preliminary ML results yet)
- **LPBFAM predictive modeling (freshman):** FEA software predicting thermal stress/deformation; poster; won project funding
- **Structural Health Monitoring (freshman, sophomore):** FEA simulation validating readings from deployable bridge sensors; paper co-author (Springer)
- **Flexible antenna design(junior):** Designed conformal antenna for cylindrical housing; printed on Kapton via custom PCB 3D printer; successful antenna, not integrated due to sensor package redesign
- **Edge FPGA ML (sophomore/junior):** Tried power-efficient LSTM on FPGA; concluded need for more guidance; pivoted away
- **Child activity/obesity sensors(junior):** Hardware (PCB) + firmware for heart-rate + accelerometer; strong team experience; paper authorship status unclear

15) Technical Issue (Concrete Anecdote)

- **Symptom:** Data loss when running multiple experiments/scripts back-to-back
- **Root cause:** A management thread constantly polled timing for parameter changes, starving the ingest thread under load
- **Fix:** Reworked thread management; added ring buffer; eliminated drops
- **Lesson:** Deployment & data transport are part of sensing; design threading and I/O budgets first

16) Teaching & Mentoring

- **TA — EMCH 290 (Thermodynamics):** Authored the entire course (quizzes, hw, slides); taught **2 semesters** (~50 students/term)
- **TA — EMCH 368 (Mechatronics Lab):** Rewrote labs, wiring diagrams, grading process; **2 semesters**; ~100+ students overall (~60 across your sections)
- **UG mentorship (2 students):**
 - **Nathan Kirk (Senior ME, ROTC):** Data collection (~15 hours), CFAR implementation; near-term short paper

- **Samuel Canilla (Sophomore ME, ROTC):** Leads hardware meeting; building traditional tracking arm; data collection; **thousands of LOC**; aiming for a conference paper
- **Time commitment: ~10–20 hrs/week** across mentoring and research support

17) Leadership & Service

- **Organizations:**
 - Daily Gamecock
 - Opinion Editor – 08/25-present
 - Edited pieces from opinion writers
 - Develop illustration plans for each piece
 - Edit multiple guest columns
 - Multi-phase marketing plan to increase interaction with student body
 - Opinion Writer – 02/25-present
 - Writing/wrote articles for the Daily Gamecock. Focus in Tech, Politics, History, and system-level thinking.
 - Written 5 Feature pieces, multiple non-features
 - Won 4th place national award for story of the year
 - See my articles here: <https://www.dailygamecock.com/staff/jc-vaught>
 - Student Government
 - Student Body Treasurer – 03/23-03/24
 - Implemented legislation from prior term in senate.
 - Increased budget from 186k to 340k.
 - Developed a series of AI-powered tools along with other digital solutions.
 - Recruited team of 15 to manage all financial requests and helped with transition to new treasurer.
 - Student Senator – 03/24-10/24
 - Wrote weekly finance legislation allocating funding
 - Researched and wrote 'data-driven' legislative changes rewriting finance codes
 - ASME
 - Advisor – 09/23 - Present
 - Vice-President – 05/23-09/23
 - Secretary – 05/22-05/23
 - Successfully managing logistics for meetings and discussions with faculty and industry sponsors for the Mechanical Engineering department, tripling the number of meetings year over year in Fall 2022

- Led the first major remodeling of the ASME lounge in 17+ years, enhancing the space for engineering students and creating a collaborative work environment
- Designed CAD models for the Tiger Burn, ran FEA analysis, and developed a physical, to-scale model for testing, in addition to a 25% increase in engineering student involvement in the event
- Secured funding and a suitable venue for the annual Tiger Burn event, including coordinating pyrotechnics for an exciting experience
- IEEE
 - Advisor – 09/23-Present
 - President – 01/23-09/23
 - Remodeled the Lounge room for the first time since building construction in 1987 via a collaboration with the EE department and the faculty advisor
 - Tripled the number of meetings per semester year over year
 - Handled necessary paperwork to register the club, in order to acquire funding from University
- Pi Tau Sigma
 - Secretary: 05/23-05/25
 - Assist the president in transforming the club from a strictly academic and professional platform to a more interactive and social community
 - Coordinate with other club officers and members to plan and execute engaging events and activities
 - Service days with group of 4; doing work in community to help out, designed sustainable watering system for Carolina sustainable gardens – took 2 semesters.
- SAE
 - President: 11/22-03/24
 - Spearheaded the revival of the SAE club on campus by increasing the membership by 300%, securing the necessary resources, and attracting passionate automotive engineers
 - Assembled a diverse and talented team of students from various engineering majors to design a Formula Electric Car and compete in the 2024 FSAE competition for the first time in almost three decades
 - Built a strong executive team to oversee the club operations and handle the marketing and administrative tasks
 - Established the club's credibility and visibility on campus by registering it with the school and leading the design process
 - Trained new members on software tools and other relevant skills for each sub-team in the club

- Restructured the club's organization to enhance transparency and communication among different teams
- Planned and organized trips to other colleges to learn from other successful teams and improve our own performance
- SAMPE
 - Secretary: 08/22-05/25
 - Scheduled rooms and recorded notes for meetings, ensuring smooth organization of events
 - Managed marketing efforts for the club, increasing visibility and engagement
 - Assisted with the annual SAMPE beam competition, contributing to its execution
- USC Rocketry Club
 - Senior Advisor: 05/23-Present
 - Secretary: 08/22-05/23
 - Co-founded the first university-affiliated NAR insured Rocketry Club in South Carolina, successfully growing it from 0 members to 120+ active members within one semester
 - Managed logistics for marketing, venues, perishable goods, and faculty/staff discussions for meetings of 50+ people
 - Assisted with the management of executive meetings with the President of the club
 - Spearheaded an aggressive marketing push within the engineering college and other related departments, increasing student awareness and engagement in the club

18) Outreach (Done & Planned)

- **Done:** Judge for **VEX** robotics competition; designed multi-station demos for middle-school open house (electric go-kart, Tesla coil, pulley/mechanical advantage, prop cars)
- **Planned:** Local high-school outreach (not specifically Aiken); periodic workshops

19) Open Science & Governance (Generic Commitments)

- **Benchmark release:** Keep name/license details TBD; promise code + scoring + baselines + docs; leaderboard when stable
- **Privacy & ethics:** Signage at collection sites; blur faces and vessel names; encrypt raw data; role-based access; include ethics note upon release

20) Personal Background & Values

- **Origin:** Homeschooled through ~15; began college with gaps in writing/basic science; transitioned to university at 18; worked multiple jobs to cover rent; secured scholarships (Palmetto Fellows, McNair Jr. Fellows)

- **Mindset:** Discipline and persistence—scope the problem, build the toolchain, iterate until it works; mentor undergrads to provide the structure you had to invent
- **Faith:** Grounds persistence and a service ethic

21) Training Plan & Career Vision

- **Coursework (next 12–18 months):** ML/CV and controls courses; RF TBD
- **Facilities:** Routine access to boats, Furuno/Simrad radars, RGB/thermal cameras, Jetson Orin, GPU servers, Hyperion
- **5-year goal:** National lab track (**NIWC/NRL**) in maritime radar/autonomy
- **10-year goal:** Startup + part-time teaching (controls, possibly calculus)

22) KPIs (Targets)

- **Year 1:** 300+ dataset downloads; ≥ 6 external benchmark runs; mentor **3–4** UGs (1–2 coauthorships)
- **Year 2:** 800+ downloads; ≥ 12 external evaluations; mentor **6–8** UGs total; v2.0 benchmark release
- **Edge:** ≥ 30 FPS at ≤ 100 ms; accuracy within drop tolerances

23) Academics & Awards

- **GPA:** **3.66** overall
- **Selected course grades:** RF I (B+), RF II (C+), Controls (A), Signal Processing (A), FPGA (B+)
- **Scholarships/Awards:** ASME / NSBE / SHPE (college merit); two EE department merit awards; **McNair Junior Fellows** (research funding); **Palmetto Fellows** (HS merit)

24) Links (Internal Use)

- **GitHub:** radar-labeler | Machine-Learning_EMCH368 | Java-Chess-Game | MO-CAP
- **QPR talk:** “Seeing in Binary: Resolving Details from Low-Bit Sensors” (mid-September)

25) Style & Redlines for Public Docs

- Refer to comparative model as “**commercial baseline**” and optionally add (**Tocaro Blue**) in parentheses
 - **Do not** name the dataset yet
 - **Do not** name the industry partner
 - Emphasize **national security**, public safety, and workforce; avoid DEI wording per preference
 - Omit timeline in the NSF personal statement
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One-Paragraph Bio (Grab-and-Go)

JC is a first-year M.S. student at UofSC working with Dr. Yi Wang on physics-aware radar-camera perception and radar-guided auto-labeling for small maritime targets. He built the lab's radar control/GUI stack and is leading three HDR radar manuscripts (multi-gain fusion, entropy-based gain selection, pulse-diverse HDR). His dataset spans ~200k radar sweeps over SC and VA waters; he reports AP_small, recall at low FAR, and embedded latency (≤ 100 ms on Jetson Orin). JC mentors undergraduates (10–20 hrs/wk), has TA'd Thermodynamics (authored course) and Mechatronics (rewrote labs), and previously co-authored a Springer chapter in structural health monitoring. Long term, he aims for a national-lab role (NIWC/NRL) advancing maritime sensing for national security, then a startup while teaching part-time.